

Fintech and absolute intergenerational mobility: Evidence from digital finance development in China

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ABSTRACT

Over the past four decades, wealth accumulation has increasingly concentrated among a small elite, particularly in developing countries. The rise of financial technology (fintech), however, offers a powerful tool to overcome these barriers and foster financial inclusion. Using microdata, we demonstrate that fintech significantly improves absolute intergenerational mobility, which represents a 67.76 % increase in the income of poor families. Our analysis reveals that fintech alleviates liquidity constraints and enhances "market luck", lowering entry barriers for economically disadvantaged individuals to pursue entrepreneurship. Furthermore, fintech's inclusive nature leads to varied effects across different households and offspring, reducing gender inequality and supporting upward mobility for low-income families. By expanding access to financial resources, fintech emerges as a crucial driver in addressing class disparities and promoting intergenerational mobility.

1. Introduction

Over the past four decades, income concentration among the top 10 % of earners has become a prominent global trend as shown in Fig. 1. In many countries, this group's share of total income consistently exceeds 50 %. While this phenomenon is most pronounced in developed nations such as the United States, France, and Germany, it is also increasingly observed in developing regions, including China and South America (Acciari et al., 2019; Bratberg et al., 2016; Mayer and Lopoo, 2008). This growing concentration of wealth not only highlights widening economic disparities but also contributes to the persistence of intergenerational inequality, which poses a critical challenge for policymakers worldwide. The impact of this concentration is particularly stark when considering the consequences for social mobility and the opportunities available to future generations.

Intergenerational mobility is a critical measure of economic opportunity, capturing the degree to which children's economic outcomes differ from those of their parents. At its core, intergenerational mobility reflects the correlation between parental and offspring incomes. A high degree of intergenerational mobility suggests that children can largely overcome the economic disadvantages associated with their family background, whereas a low degree of mobility signals the persistence of parental economic status. This persistence—where "the poor stay poor"—is often interpreted as evidence of a rigid class structure and limited social mobility. Such concerns have been extensively

documented in the literature, which emphasizes that low intergenerational mobility not only perpetuates economic inequality but also poses significant challenges for social fairness and long-term economic dynamism (Bavaro and Patriarca, 2022; Chetty et al., 2017, 2014; Deutscher and Mazumder, 2023; Solon, 1992).

Financial technology (fintech) has emerged as a transformative force in the global financial landscape, offering potential solutions to the persistent issues of inequality and limited mobility. Fintech encompasses innovations such as mobile payments, online lending, and digital investment platforms. It is characterized by two key features: digitalization and inclusivity. Digitalization reduces the cost of financial services and improves accessibility (Goldfarb and Tucker, 2019; Hua and Huang, 2021; Molla and Biru, 2023). Inclusivity refers to fintech's ability to extend financial services to previously underserved populations, potentially mitigating financial exclusion and inequality. Given these features, fintech holds significant promise as a tool for enhancing intergenerational mobility by improving access to financial resources and reducing barriers that hinder economic opportunity. This paper explores the potential of fintech to contribute to reducing intergenerational inequality by improving the economic opportunities available to disadvantaged families.

Despite its potential, the literature on fintech's impact on absolute intergenerational mobility remains limited. While there is growing evidence of fintech's ability to reduce financial exclusion and facilitate economic opportunities, studies on its specific role in improving

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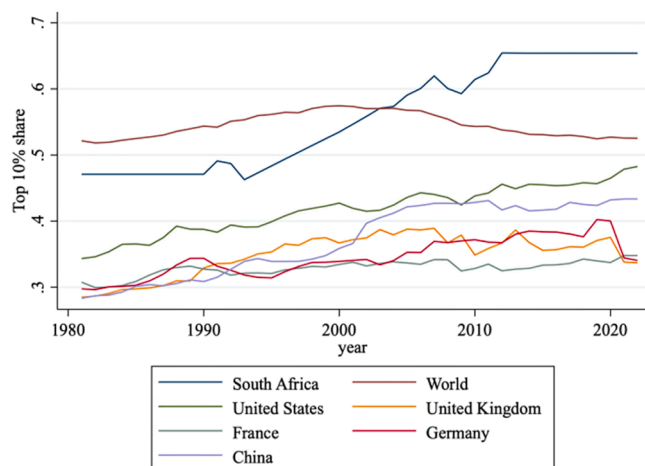


Fig. 1. The top 10 % income share.

Note: This figure plots the income share of the top 10 % among some countries from 1980 to 2020. The data is from world inequality database.

intergenerational mobility are still scarce (Yang et al., 2024). This paper seeks to address this gap by examining the impact of fintech on intergenerational mobility in China. As the world's largest developing economy, China has seen particularly rapid growth in fintech (Hau et al., 2019; Yang and Zhang, 2022). Platforms like Alipay, with over one billion active users, have become key drivers of digital financial inclusion. China's fintech ecosystem not only spans vast regions but also exhibits a higher degree of inclusivity than those in many developed countries (Hua and Huang, 2021; Molla and Biru, 2023). At the same time, other developing countries do not have detailed data on father-son income to explore the intergenerational mobility of income (Asher et al., 2024). However, there are detailed micro surveys in China, such as the China Family Panel Studies (CFPS), which provide data support for us to measure intergenerational income mobility (Fan et al., 2021). Therefore, understanding how fintech influences intergenerational mobility in China can offer valuable insights for both developing nations and developed economies seeking to harness fintech's inclusive potential.

The literature on intergenerational mobility has primarily focused on various structural factors that influence economic outcomes across generations, such as human capital investment, family characteristics, and policy interventions. The classical framework proposed by Becker and Tomes (1979, 1985) suggests that children's economic outcomes are largely shaped by parental investments in education and other forms of capital, as well as by "market luck". Importantly, "market luck" is not determined by families or individuals but is shaped by broader macro-economic and market factors. We define that "market luck" is an important measure of fairness in the external environment, particularly with regard to equal opportunities in the labor market.

Building on this framework, researchers have studied factors affecting intergenerational mobility, including income distribution, family characteristics, and policy interventions. Regarding income distribution, many studies focus on wealth transmission across generations. For example, Bjorklund et al. (2012) found that wealth transfers are a key factor in maintaining wealth among the rich, based on Swedish father-son pair data. Landersø and Heckman (2017) demonstrated that redistributive policies, such as transfer payments, contribute to high intergenerational mobility in Denmark. Other research examines family-related factors, such as gender, family size, and assortative mating in marriage (Chetty et al., 2014; Davis and Mazumder, 2020; Mitnik et al., 2015). From a policy perspective, studies have analyzed the impact of events like the Nordic oil boom (Bütikofer et al., 2018), school reforms (Duan et al., 2022; Nybom and Stuhler, 2015; Pekkarinen et al., 2009), and government spending (Huang et al., 2021). Despite these contributions, the role of fintech in absolute intergenerational

mobility remains underexplored, existing a research gap that this study seeks to address.

Recent studies on fintech have presented divergent views on its impact on inequality. Some argue that fintech exacerbates inequality by deepening the rural-urban divide (Liu et al., 2020) or disproportionately benefiting high-income countries (Demir et al., 2020; Hua and Huang, 2021). Others highlight fintech's inclusive potential, suggesting that it fosters entrepreneurship, enhances financial inclusion, and reduces economic disparities (Hua and Huang, 2021; Molla and Biru, 2023). However, these studies primarily focus on intra-generational inequality and often overlook its impact on intergenerational inequality. This oversight is significant because it is challenging to fully isolate the effects of intergenerational inequality from the overall inequality observed within a generation. The exclusion of intergenerational mobility may lead to inconsistent or incomplete conclusions regarding the broader social impact of fintech. To address this gap, we focus on the effect of fintech on intergenerational inequality, specifically through the lens of intergenerational mobility.

Firstly, we measure the absolute intergenerational mobility by the method of upward rank mobility ("URM") (Bhattacharya and Mazumder, 2011). Unlike prior studies that primarily focus on relative mobility and intra-family capital transmission (Becker and Tomes, 1979; Chetty et al., 2014), this approach not only captures the correlation between parental and offspring incomes but also directly assesses the actual economic gains achieved by offspring. Our estimates indicate that the $URM_{50,20}^1$ value in China is 32 %, which is lower than in many developed countries.

Next, we investigate the impact of fintech on absolute intergenerational mobility in China. Our results reveal that a 1 % increase in the fintech index is associated with an average 5.3 % increase in absolute intergenerational mobility, which corresponds to a 67.76 % increase in income for low-income families.

Furthermore, we examine the channels through which fintech influences absolute intergenerational mobility. Our findings indicate that fintech significantly alleviates liquidity constraints for both parents and children, thereby enhancing parental investment in education and facilitating entrepreneurial activities among disadvantaged offspring. In addition, fintech improves "market luck" by boosting economic vitality, and expanding internet access.

Finally, we highlight the inclusive nature of fintech. Our results demonstrate that fintech's impact varies across offspring and household characteristics, revealing that it reduces gender disparities and fosters upward mobility, particularly for low-income families.

Our study contributes to the literature in three key ways. First, it addresses a significant research gap regarding the impact of fintech on absolute intergenerational mobility. While much of the existing literature on intergenerational mobility primarily focuses on the factors influencing relative intergenerational mobility (Bjorklund et al., 2012; Chetty et al., 2014; Davis and Mazumder, 2020; Landersø and Heckman, 2017; Mitnik et al., 2015), it largely overlooks how these factors affect absolute intergenerational mobility. Importantly, factors influencing relative mobility do not necessarily affect absolute mobility. An increase in relative intergenerational mobility does not always correspond to an increase in absolute intergenerational mobility (Berman, 2022; Deutscher and Mazumder, 2023). Thus, our study fills this gap by focusing on the effect of fintech on absolute intergenerational mobility.

Second, this study provides new evidence on fintech's role in influencing income inequality, offering a fresh perspective. The existing literature on fintech's impact on income inequality remains divided. However, these studies predominantly focus on intra-generational inequality (Gao et al., 2022; Liu et al., 2020; Sun and Tang, 2022), neglecting the important issue of intergenerational inequality. This

¹ $URM_{50,20}$ conditions on families below the median and the likelihood that they exceed their parents' rank by >20 percentiles.

oversight may contribute to the inconsistent conclusions found in the literature. Our study complements the existing research by examining fintech's impact on intergenerational inequality, specifically through the lens of absolute intergenerational mobility.

Third, we identify additional channels through which fintech influences intergenerational mobility, such as "market luck." In our study, we define "market luck" as the fair opportunities available in the labor market. Previous literature has largely focused on the micro-level impact of fintech on liquidity constraints (Huang et al., 2021; Lou and Li, 2022; Yang et al., 2024), often overlooking its macro-level impact on the fairness of the external environment. At the same time, the research on intergenerational mobility is often limited to the perspective of social capital and human capital (Bjorklund et al., 2012; Chetty et al., 2014; Davis and Mazumder, 2020; Landersø and Heckman, 2017; Mitnik et al., 2015), but ignores the role of "market luck," namely, fair labor market opportunities. Our study contributes to the literature by exploring fintech's role in enhancing inclusion at the macro level, particularly through its effect on improving "market luck."

The remainder of the paper is organized as follows: Section 2 presents the theoretical framework. Section 3 discusses the index, the data and methodology. Section 4 presents the baseline results. Section 5 provides robustness checks. Section 6 explores the mechanisms through which fintech enhances intergenerational mobility. Section 7 offers a heterogeneity analysis, and Section 8 concludes.

2. Theoretical analysis

Based on Becker and Tomes (1979, 1985), we define the utility function of parents as follows:

$$U_0 = (1 - \alpha) \ln C_0 + \alpha \ln Y_1 \quad (1)$$

Here, U_0 represents the parents' utility, C_0 is their consumption, and Y_1 is the child's income. The parameter α indicates the importance parents place on their child's welfare; a higher α implies greater attention to the child.

The children's income depends on their education level and market conditions:

$$\ln Y_1 = \mu_1 + p h_1 \quad (2)$$

where μ_1 captures "market luck", h_1 is the child's education level, and p is the return on education.

The education level is a function of parental investment:

$$h_1 = \theta \ln(I_0) \quad (3)$$

Here, I_0 is the parent's investment in the child's education. Parents allocate their income between their own consumption and education investment, subject to the following budget constraint:

$$C_0 + I_0 = \tau_0(Y_0) \quad (4)$$

where τ_0 reflects liquidity constraints. A lower τ_0 indicates stricter constraints, which reduce the parents' disposable income:

$$\frac{\partial \tau_0}{\partial Y_0} < 0 \quad (5)$$

Maximizing parental utility yields the optimal educational investment:

$$I_0^* = \frac{\alpha \tau_1 p \tau_0(Y_0)}{1 - \alpha + \alpha \tau_1 p} \quad (6)$$

Fintech, as a financial service, reduces liquidity constraints:

$$\frac{\partial \tau_0}{\partial D} < 0 \quad (7)$$

where D represents the level of fintech. Substituting this into Eqs. (4); (5); and (6) shows:

$$\frac{\partial I^*}{\partial D} > 0 \quad (8)$$

Thus, fintech alleviates liquidity constraints for low-income families, enabling greater investment in children's education. This improves the child's education level and income, fostering upward intergenerational mobility. The ratio of the child's income to the parent's income measures intergenerational mobility:

$$\text{Mobility} = \frac{Y_1}{Y_0} = \frac{e^{\mu_1} + e^{p\theta} \times \frac{\alpha p \tau_0(Y_0)}{1 - \alpha + \alpha p}}{Y_0} \quad (9)$$

Given Eqs. (4), (7), and (8), it follows that:

$$\frac{\partial Y_1}{\partial D} > 0 \quad (10)$$

$$\frac{\partial \text{Mobility}}{\partial D} > 0 \quad (11)$$

In short, as fintech reduces the liquidity constraints faced by low-income parents, they can increase their investment in their children's education, extending the duration of schooling and improving the children's human capital. This, in turn, enhances the children's income relative to their parents, achieving upward intergenerational mobility. Thus, the first hypothesis of this paper is proposed:

H1: Fintech promotes upward intergenerational mobility by reducing parental liquidity constraints and increasing investments in children's education.

Fintech also complements traditional finance, enhancing capital flow and addressing liquidity constraints for low-income families. In rural areas, where agricultural income is low and opportunities are limited, fintech provides the necessary capital for non-agricultural employment and income diversification. For low-income households, which face barriers to entrepreneurship due to unclear assets and risk-averse lenders (Agarwal and Hauswald, 2008; Hoskisson et al., 2000; Welter and Smallbone, 2011), fintech reduces information asymmetry and improves access to funding. This promotes entrepreneurship, enhances human capital, and fosters upward mobility.

H2: Fintech alleviates liquidity constraints for offspring, promoting non-agricultural employment, entrepreneurship in low-income families, and innovation in small and micro enterprises, thereby enhancing intergenerational mobility.

Fintech also influences intergenerational mobility through "market luck". Historically, financial resource scarcity in underdeveloped regions has led to talent outflows and negative cycles. By unifying capital markets and supporting entrepreneurship, fintech revitalizes these regions, increases job opportunities, and boosts wages, enabling children to surpass their parents' income.

Additionally, fintech narrows the digital divide, enhancing internet access and reducing information asymmetry. This fosters "learning by doing" through accessible online resources, enabling children to acquire human capital and achieve upward mobility.

H3: Fintech promotes upward generational mobility by enhancing "market luck" through urban entrepreneurship and increased internet access.

In summary, fintech—through its inclusive and digital nature—reduces liquidity constraints and improves "market luck", fostering intergenerational mobility as shown in Fig. 2. This includes increased investment in education, support for non-agricultural employment, encouragement of entrepreneurship in disadvantaged families, and innovation in small enterprises.

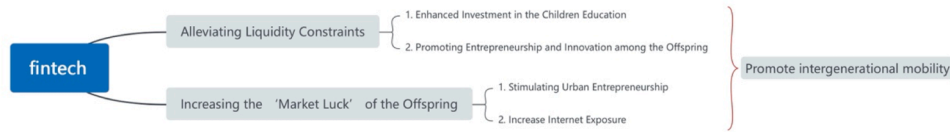


Fig. 2. The analysis framework diagram.

3. Index, data and model

3.1. Calculation of absolute intergenerational mobility

We use the method of upward rank mobility (“URM”) to measure absolute intergenerational mobility. The reason why we choose absolute intergenerational mobility rather than relative intergenerational mobility to measure intergenerational equity is that absolute intergenerational mobility can better reflect the changes of children relative to their parents, which can better explain the welfare changes. Better relative intergenerational mobility does not necessarily mean better absolute intergenerational mobility, that is, poorer families get richer.

The method of upward rank mobility (“URM”) calculates the probability that a child with a parent’s income in the m % quartile is greater than the parent’s income rank of n %, such as the formula (12):

$$URM_{m,n} = P(r_1 - r_0 \geq n\% | r_0 \leq m\%) \quad (12)$$

Where r_1 is the income rank of the offspring and r_0 is the income rank of the parent. This approach has a number of advantages: first, measuring income by ranking can solve the problem of income comparability between parents and children and alleviate measurement error (Nyblom and Stuhler, 2015; O’Neill et al., 2007). Secondly, limiting the parental ranking, such as limiting it to the bottom 50 %, can be used to study the poor. Finally, this method can study the response of children, while the previous literature cannot examine the impact on children by using the intergenerational elasticity method of relative intergenerational mobility (Deutscher and Mazumder, 2023).

Based on this approach, we calculate $URM_{50,20}$ for China where the parent’s income is ranked in the bottom 50 % and the child’s income is ranked 20 % above the parents. We estimate the $URM_{50,20}$ at 32 % in China, compared with 47 % in Australia (Deutscher and Mazumder, 2023), 45 percent in the U.S., 45.2 percent in Sweden and 45.7 percent in Canada (Corak et al., 2014). It’s smaller than developed countries, but it’s similar to other estimation for China (Fan et al., 2021).

3.2. Data

The data used in this paper mainly includes three parts: The first is the micro-database of the China Family Panel Studies (CFPS), which collects data at the three levels of individual, family and community through tracking to reflect the changes in China’s society, economy, population, education and health (Xie and Hu, 2014), which provides a data basis for this paper to study intergenerational mobility from a micro-perspective. Among them, the sub-indicators of digitalization better represent the development level of fintech in various regions of China (Guo et al., 2020), and this paper uses the sub-indicators for regression. The third is the data from the City Statistical yearbook, which is used as a control variable at the regional level to exclude systematic differences caused by regions.

3.2.1. China family panel studies

The microdata used in this article come from the China Family Panel Studies (CFPS). The CFPS is a nationwide, comprehensive social tracking survey. It aims to collect data at the individual, household, and community levels. It reflects social, economic, demographic, educational, and health changes in China (Xie and Hu, 2014).

The CFPS focuses on the economic and non-economic well-being of

the Chinese population. It includes a wide range of research topics such as economic activity, educational attainment, family relationships and family dynamics, population migration, and physical and mental health. In 2010, the CFPS conducted a baseline survey in 25 provinces/municipalities/autonomous regions across China, covering 14,960 households and 42,590 individuals. All household members defined in the baseline survey and their future newborn blood/adopted children are defined as CFPS genetic members and are permanently tracked in the CFPS survey, which is visited every two years. This provides the database for this article to study absolute intergenerational mobility from a micro perspective. To be consistent with the Fintech Index, this article uses data from the China Family Panel Studies in 2012, 2014, 2016, 2018 and 2020.²

3.2.2. Fintech index

This study relies on the Peking University Digital Financial Inclusion Index of China (PKU_DFIIC), developed in collaboration with the Digital Finance Research Center of Peking University, the Shanghai New Finance Research Institute, and Ant Financial Services Group (Guo et al., 2020). This index utilizes micro-level data on fintech sourced from Ant Financial Services, a leading internet financial institution in China.

The PKU_DFIIC is designed to capture the multifaceted aspects of fintech through its comprehensive framework. It assesses digital inclusive finance across four key dimensions: mobility, affordability, credit-worthiness, and convenience. By integrating traditional inclusive finance indicators with the specific characteristics of digital financial services, the index provides robust insights into the impact of fintech.

Spanning from 2011 to 2020, the PKU_DFIIC offers city-level data that enable longitudinal and cross-sectional analyses. Its hierarchical analysis methodology ensures reliable and comparable measurements, making it an ideal explanatory variable for studying the effects of fintech on intergenerational mobility.

3.3. Model

In this article, we use a panel fixed effects regression model to construct the following regression equation (Duan et al., 2022; Feng and He, 2022; Yang and Qiu, 2016; Yang et al., 2024):

$$URM_{50,30} = \beta_0 + \beta_1 Fintech_{c,t-1} + \gamma c_{ict} + \theta m_{ct} + \delta_c + \epsilon_t + \epsilon_{ict} \quad (13)$$

The explanatory variable $URM_{50,30}$ is a index to measure absolute intergenerational mobility. $URM_{50,30}$ is 1 when the parent’s income is in the bottom 50 percent of the peer group, but the child’s income rank is 30 per cent above the parent’s income rank, otherwise is 0. The core explanatory variable $Fintech$ is the city-level PKU_DFIIC index. To mitigate the problem of reverse causality, this article uses the logarithm of the fintech index with a one-period lag as the core explanatory variable.

At the same time, the article controls for variables at the individual level, family level and city level (Yang et al., 2024). At the individual level, we control age, age squared, education, household registration, and gender of offspring. At the family level, we control per capita income, household size, and parents’ education. At the city level, we control economic level, education level, traditional financial

² The analysis of the data of restricted districts and counties in this article was carried out in the computer room of the China Social Sciences Survey Center of Peking University.

development level, population and school. Additionally, we control for year fixed effects and city fixed effects to control other omitted variables that do not vary with district or time.

Some studies suggest that income around the age of 30 can serve as a reliable proxy for permanent income (Aaronson and Mazumder, 2008; Baker and Solon, 2003; Grawe, 2006; Lefranc, 2018; Mazumder, 2005). Following this approach, we focus on offspring aged between 20 and 40 years old, censoring individuals outside this range to ensure consistency in the analysis. This results in a final sample size of 4693 observations as shown in Table 1.

4. Baseline regression

The regression results based on Eq. (13) are presented in Table 2. To ensure the robustness of our findings, we adopt a stepwise approach that incrementally includes control variables. Specifically, column (1) excludes any level control variables, column (2) includes individual and household-level control variables, and column (3) incorporates control variables at the individual, household, and city levels. In all columns, the coefficients for the core explanatory variables are significantly positive, confirming the robustness of our results.

In particular, column (3) shows that even after controlling for individual, household, and city-level variables, the coefficients remain positive and statistically significant at the 5 % level. These results provide strong evidence that fintech promotes absolute intergenerational upward mobility, underscoring its inclusive impact. Economically, a 1 % increase in the fintech index is associated with an average 5.3 % increase in the level of absolute intergenerational mobility.

To further illustrate the welfare impact, we convert these results into monetary terms. According to our data, the average income of offspring from families in the bottom 50 % of the income distribution is 33,879 yuan, while offspring from families in the 30 % to 80 % income range earn an average of 52,319 yuan. Therefore, a 1 % increase in the fintech index corresponds to an increase of at least 977 yuan per household. From 2012 to 2020, the fintech index increased by 188 %. Consequently, fintech can raise the income of households in the bottom 50 % by at least 22,959 yuan annually. This represents a 67.76 % increase in the income of poor families.

5. Robustness test

For the robustness of the results, we do some robustness tests. Firstly, the variable measurement problem is tested. We use different ways to measure intergenerational mobility and fintech. Secondly, we narrow the sample to exclude the influence of abnormal samples. For example,

Table 1
Summary statistics.

VARIABLES	Number	Mean	Std	Min	Max
URM _{50,30}	4693	0.202	0.401	0	1
Fintech	4693	4.694	0.804	2.058	5.744
Age	4693	26.441	4.75	20	40
Age ²	4693	721.696	270.088	400	1600
Education	4693	3.881	1.302	0	8
Rural	4693	0.664	0.473	0	1
Sex	4693	1.097	1.278	0	5
Household size	4693	0.203	0.403	0	1
Parents' education	4693	4.398	2.047	1	16
Economic level	4693	16.83	1.001	14.619	19.774
Education level	4693	0.035	0.022	0.009	0.148
Traditional financial development level	4693	1.057	0.696	0.168	7.450
Population	4693	6.143	0.534	4.516	8.130
School	4693	6.463	0.811	3.555	8.29

Note: Due to the confidentiality of personal information geographical location, the maximum and minimum values at the city level are based on the data before matching, rather than the data after matching.

Table 2

Estimated effects of the fintech on absolute intergenerational mobility.

VARIABLE	(1) URM _{50,30}	(2) URM _{50,30}	(3) URM _{50,30}
Fintech	0.034* (0.020)	0.039* (0.021)	0.053** (0.021)
Individual/Family-level Controls	No	Yes	Yes
City-level Controls	No	No	Yes
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observation	4693	4693	4693
R-squared	0.032	0.073	0.074

Note: This table reports the effects of the fintech on absolute intergenerational mobility. Column (1) doesn't have any control variables. Column (2) controls individual and family-level variables. Column (3) controls individual, family and city variables. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

we narrow the sample year to exclude the influence of the epidemic. Finally, the possible endogeneity problem is tested. We add the ex-ante trend term at the regional level and use the instrumental variable method to solve the possible omitted variable problem. All the results prove robustness of baseline regression.

5.1. Replacement of independent variables

The explained variable is the absolute intergenerational mobility measured by the upward rank mobility. The disadvantage of its way is that the value is subjective. Therefore, we adopt other methods to measure absolute intergenerational mobility and change the subjective value to test the robustness. Moreover, it is supported by relative intergenerational mobility. The results are presented in Panels A and B in Table 3, respectively.

In Panels A and B, we use upward rank mobility method of absolute intergenerational mobility as the baseline. But we change the rank difference between the offsprings and the parents. The dependent variables in columns (1), (2), (3), (4) and (5) are whether the offspring's income rank is greater than 0 %, 10 %, 20 %, 25 %, and 40 % of the parent's income rank when the parent's income is ranked in the bottom 50 %, respectively. By changing the rank difference, we can know how big the effect of fintech on absolute intergenerational mobility is. The results show that the coefficient of fintech is all significantly positive, which proves robustness.

In Panel B, we change the parents' rank. The dependent variables in columns (1), (2), (3), and (4) are the parent's income rank is smaller than 20 %, 30 %, 40 % or greater than 50 %, respectively. We find that by moving fathers further down the income ladder, accounting for poorer households, fintech still promotes absolute upward intergenerational mobility. And for rich households, where the parent is in the top 50 percent of income earners, we find no positive effects of fintech. This illustrates the robustness of the results.

In Panel C, we calculate relative intergenerational mobility. The dependent variable in column (1) is whether the child whose father is in the bottom 25 % of income is in the top 75 %. The coefficient is still significantly positive. Therefore, the results also prove the robustness.

5.2. Eliminate some samples

To prevent the influence of abnormal samples, we eliminate part of the samples. We eliminate the 2020 data to prevent the impact of the pandemic. This is because we allow for the possibility that the pandemic may have reduced earnings significantly in 2020, thus causing our estimates of absolute intergenerational mobility to be inaccurate. The results are shown in Table 4. The coefficient of fintech remains significantly positive at the 5 % significance level. It indicates that after we

Table 3
Replacement of dependent and independent variables.

VARIVABLE	Panel A: Change the rank gap between children and parents				
	(1) $URM_{50,0}$	(2) $URM_{50,20}$	(3) $URM_{50,20}$	(4) $URM_{50,25}$	(5) $URM_{50,40}$
Fintech	0.156*** (0.017)	0.156*** (0.017)	0.146*** (0.015)	0.060** (0.025)	0.050*** (0.019)
Observation	4693	4693	4693	4693	4693
R-squared	0.095	0.095	0.084	0.082	0.069

VARIVABLE	Panel B: Change the parent ranking			
	(1) $URM_{20,30}$	(2) $URM_{30,30}$	(3) $URM_{40,30}$	(4) $URM_{50+,30}$
Fintech	0.038* (0.021)	0.046** (0.021)	0.054*** (0.020)	0.004 (0.002)
Observation	4693	4693	4693	4693
R-squared	0.068	0.071	0.073	0.025

Panel C: Relative intergenerational mobility	
VARIVABLE	(1) Relative intergenerational mobility
Fintech	0.018* (0.010)
Observation	4693
R-squared	0.034

Note: This table reports the results of replacement of dependent and core independent variable. In Panels A and B, we use upward rank mobility method of absolute intergenerational mobility as the baseline. But we change the rank difference between the offsprings and the parents in Panel A. The dependent variables in columns (1), (2), (3), (4) and (5) are whether the offspring's income rank is greater than 0 %, 10 %, 20 %, 25 %, and 40 % of the parent's income rank when the parent's income is ranked in the bottom 50 %, respectively. In Panel B, we change the parents' rank. The dependent variables in columns (1), (2), (3), and (4) are the parent's income rank is smaller than 20 %, 30 %, 40 % or greater than 50 %, respectively. In Panel D, we calculate relative intergenerational mobility. The dependent variable in column (1) is whether the child whose father is in the bottom 25 % of income is in the top 75 %. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Table 4
Eliminate some samples.

VARIVABLE	(1) $URM_{50,30}$
Fintech	0.053** (0.021)
Observation	4693
R-squared	0.074

Note: This table reports the results of excluding the impact of pandemic. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

eliminate some abnormal sample, the results are still robust.

5.3. Endogenous treatment

To solve the problem of missing variables, we control policy shocks, add ex-ante trend items and instrumental variables. The results are

Table 5
Solve the endogeneity problem.

VARIVABLE	Panel A: Add time pre-trend terms		
	(1) $URM_{50,30}$	(2) $URM_{50,30}$	(3) $URM_{50,30}$
Fintech	0.049** (0.022)	0.049** (0.022)	0.049** (0.022)
City-level pre-trend $\times$ Year	YES	YES	YES
City-level pre-trend $\times$ Year ²	NO	YES	YES
City-level pre-trend $\times$ Year ³	NO	YES	YES
Observation	4693	4693	4693
R-squared	0.074	0.074	0.074

Panel B: IV method			
VARIVABLE	(1) First stage Fintech	(2) Second stage $URM_{50,30}$	(3) GMM $URM_{50,30}$
	Fintech	0.209** (0.095)	0.209** (0.095)
IV	−0.080*** (0.017)		
Cragg-Donald Wald F statistic	435.76		
Kleibergen-Paap Wald rk F statistic	22.27		
Observation	4693	4693	4693
R-squared	0.942	0.037	0.037

Note: This table reports the results of solving the endogeneity problem. In Panel A, we add time pre-trend terms to control omitted variables. In Panel B, we use IV method. The results in columns (1) and (2) are the first stage and the second stage. And we use GMM method to estimate in column (3). All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

shown in Table 5. In Panel A, we add time pre-trend terms to control omitted variables. The results are still significantly positive.

In Panel B, we use instrumental variables (IV) methodology. Specifically, it utilizes the degree of topographic relief as an instrumental variable (Ivus and Boland, 2015). Topographic relief is exogenous and determined by geographic factors, making it unrelated to current absolute intergenerational mobility but potentially influencing fintech development. More topographic relief suggests more challenging terrain for fintech development.

Although we have controlled the city fixed effect and the variables at the city level blocking other channels through which the terrain relief affects intergenerational mobility (Acconcia et al., 2014), there is still the possibility that the degree of topographic relief affects intergenerational mobility through other channels. Since the higher the degree of topographic relief, it is often associated with unfavorable development conditions (Nunn and Puga, 2012) and then affects the intergenerational mobility of the residents. Thus, omitting these variables will lead to an underestimation of the results (Acemoglu et al., 2001). As shown in Column 1 of Appendix Table A2, the coefficients obtained by the instrumental variable method without the control variables are lower, proving that the omission of variables will lead to the underestimation of the results. Our results are still significantly positive despite underestimation, which indicates the credibility of the results.

To further prove the exogeneity of topographic relief, we run regression discontinuity on the intergenerational mobility level based on the median of topographic relief as shown in Fig. A1 in appendix. Based on the median of fintech, we classify those with a fintech level below the median as the control group, and those with a fintech level above the

median as the experimental group. We can find that the degree of topographic relief has a significant jump mainly in the experimental group, which indicates that the degree of topographic relief mainly affects intergenerational mobility through fintech. This simply indicates the exogeneity of topographic relief.

We also conduct a placebo test. If the topographic relief satisfies exogeneity, that is, exogeneity can only influence intergenerational mobility through fintech. To verify this conjecture, we use microdata of rural individuals from the 1988 China Residents' Income Survey Project (CHIP). This is because China's fintech originated from the launch of Alipay in 2004. We ensure that rural areas in 1988 were not affected by the development of fintech. So we use this data to examine whether the topographic relief will affect the absolute intergenerational mobility. If it does not, then it proves our conjecture. The results are shown in Table A1 of the appendix. The topographic relief has no significant effect on the absolute intergenerational mobility, even if we reduce the magnitude of the absolute intergenerational mobility. This verifies the exogeneity of topographic relief.

Following the approach outlined in Nunn & Qian (2014), we use the number of Chinese netizens in a previous period to ensure the instrumental variable is not absorbed by city fixed effects. This approach leverages the interaction between the instrumental variable (topographic relief) and the number of netizens in a two-stage least squares (2SLS) regression framework.

The results in Panel B of Table 5 are presented as follows: Column (1) reports the first stage results, showing that the coefficient of the instrumental variable (topographic relief) is significantly negative at the 1 % level. This indicates that topographic relief indeed affects fintech development as expected, confirming the exogeneity of the instrumental variable. Moreover, the F-statistic is greater than ten, and the results pass the weak instrumental variable test.

Column (2) presents the second stage results. Here, the fitted value for fintech is significantly positive at the 1 % level, even after accounting for endogeneity using the instrumental variable approach. This suggests that fintech development continues to significantly enhance absolute intergenerational upward mobility. In addition, in the column (3), we use the dynamic panel regression model, and the results do not change at all.

Finally, we also adopt the degree of topographic relief and the impact of the "Broadband China" policy as instrumental variables. "Broadband China" is a policy issued by China in 2013 to build the public infrastructure of broadband networks. Since "Broadband China" is an exogenous policy issued by China, it satisfies its exogenous nature. In addition, the broadband infrastructure brought about by "Broadband China" is closely related to the development of fintech, so it meets the requirements of relevance. This result is shown in the column (2) of Table A2 in the appendix and is consistent with the previous one. This indicates the robustness of the result.

These findings not only validate the robustness of the instrumental variable approach in addressing endogeneity concerns but also underscore the positive impact of fintech on absolute intergenerational mobility. By carefully addressing potential sources of bias, such as reverse causation and omitted variables, the study strengthens the credibility of its conclusions regarding the inclusive benefits of fintech advancements.

6. Mechanism

It has already been demonstrated that fintech promotes intergenerational upward mobility. Next, based on the Becker and Tomes (1979, 1985), we try to find its channel. Because of its digital and inclusive nature, fintech can alleviate liquidity constraints and increase market luck.

6.1. Alleviating liquidity constraints

Fintech, as a financial service, plays a crucial role in alleviating liquidity constraints. Capital is a significant factor in intergenerational mobility, as highlighted by Becker and Tomes (1979, 1985) in their model of human and traditional capital. They argue that borrowing constraints hinder mobility for low-income families. Fintech addresses this issue in two primary ways: Firstly, it alleviates liquidity constraints for parents, enabling increased investment in their children's education. The family investment in education plays an important role in intergenerational mobility (Yang and Qiu, 2016). Low-income families often reduce investment in their children's education due to financial constraints, and even let their children drop out of school to farm. It undoubtedly further expands income inequality and leads to downward intergenerational mobility. Therefore, when fintech alleviates the financing constraints of parents, it can increase the investment in children's education. This will increase the permanent income of the children, which will improve the income ranking of the children and achieve upward intergenerational mobility (Neidhöfer et al., 2018).

Secondly, it empowers younger generations by easing their liquidity constraints. Fintech can supplement traditional finance and broaden access to financial services for poor people (Hua and Huang, 2021). This democratization of capital reduces concentration among a few segments of society, enabling micro and small businesses as well as low-income groups to access financial services. This democratization not only facilitates economic mobility but also fosters innovation and entrepreneurship among younger generations (Agarwal and Hauswald, 2008; Hoskisson et al., 2000; Welter and Smallbone, 2011). Entrepreneurship enables offspring to obtain higher income and change the pattern of income distribution, thus achieving upward intergenerational mobility (Quadriini, 1999, 2000). Therefore, fintech plays a pivotal role in promoting intergenerational mobility by mitigating constraints that hinder both parental and offspring generations.

6.1.1. Enhanced investment in the children education

Fintech alleviates the mobility constraints of the father's generation so that the father's generation strengthens its investment in the education of the offspring. So, we empirically analyze whether fintech can promote the education level of the offspring. To highlight the inclusion of fintech, we run grouped regressions based on the median parental educational attainment. As a result, the regression results are obtained as shown in Table 6.

Columns (1), (2) and (3) show the regression results for the full sample, low paternal education and high paternal education groups, respectively. The results in column (1) show that fintech increases the education level of the offspring at the 10 % significance level. It proves that fintech eases the liquidity constraint of the parent generation and makes the parent generation strengthen its investment in the education

Table 6
Enhanced investment in the children education.

	(1) Full sample	(2) Low level of parents' education	(3) High level of parents' education
VARIABLE	Educational level of offspring	Educational level of offspring	Educational level of offspring
Fintech	0.298* (0.174)	0.192* (0.106)	−0.001 (0.108)
Observation	4693	1924	2752
R-squared	0.349	0.273	0.363

Note: This table reports the results of whether fintech enhances investment in the children education. Column (1) uses full sample. We run grouped regressions based on the median parental educational attainment as shown in Columns (2) and (3). All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

of the offspring. The post-grouping results show that the coefficient of fintech is significantly positive at the 10 percent significance level for the low education group of the paternal generation, but the coefficient is not significant for the high education group of the paternal generation. This suggests that the inclusive nature of fintech acts more on the low paternal education group, reducing the intergenerational education gap and achieving intergenerational upward mobility.

6.1.2. Promoting entrepreneurship and innovation among the offspring

Fintech plays a pivotal role in alleviating liquidity constraints, particularly beneficial for micro and small enterprises as well as economically disadvantaged families. It fosters inclusivity by providing convenient financial services and lowering financing costs (Hua and Huang, 2021). Therefore, fintech enhances entrepreneurial opportunities and innovation among offspring (Bertoni et al., 2022; Welter and Smallbone, 2011; Yang et al., 2024; Ye et al., 2022). We examine fintech's impact across three dimensions: non-farm employment of rural people, entrepreneurship, and innovation of micro and small enterprises. It analyzes how fintech inclusively mitigates poverty among offspring, thereby preventing a return to poverty.

Non-farm employment is defined as a shift from agricultural to non-farm work. Fintech facilitates this transition by offering rural populations accessible financing channels, thereby diversifying income sources and promoting upward mobility. The results are shown in Panel A of Table 7. In column (1), it shows that fintech has a positive effect on non-farm employment of rural sample. It underscores fintech's role in overcoming financial barriers to non-farm employment, crucial for achieving intergenerational upward mobility. Column (2) introduces a cross-multiplier term of fintech and household income. The coefficient

Table 7
Promoting entrepreneurship among the offspring.

Panel A: Non-farm employment		
	(1)	(2)
VARIVABLE	Rural sample Non-farm employment	Interaction terms Non-farm employment
Fintech	0.020** (0.008)	0.136*** (0.032)
Fintech × ln(income)		−0.010*** (0.003)
Observation	3105	2864
R-squared	0.053	0.059

Panel B: Entrepreneurship					
	(1)	(2)	(3)	(4)	(5)
VARIVABLE	Interaction terms	25 %	50 %	75 %	100 %
Fintech	0.323*** (0.084)	0.032* (0.019)	0.009 (0.018)	−0.045 (0.033)	−0.042 (0.083)
Fintech × ln (income)	−1.125* (0.573)				
Observation	4321	1066	1074	1071	1442
R-squared	0.086	0.167	0.164	0.197	0.198

Note: This table reports the results of whether fintech promotes non-farm employment and entrepreneurship in the offspring. In Panel A, we explore the effect of fintech on non-farm employment of rural offspring. Column (2) adds interaction terms based on column (1). In Panel B, we find how fintech influence entrepreneurship. Columns (2), (3), (4) and (5) are grouped regression according to total household income. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

of fintech is still significantly positive and the term is significantly negative at the 1 % significant level. It indicates that fintech can alleviate liquidity restrain to facilitate non-farm employment.

Entrepreneurship historically faced challenges due to limited and costly traditional loans. Fintech leverages digital tools to reduce information asymmetry, enhance fund flow, and accelerate capital turnover, thus supporting entrepreneurial ventures. The results are shown in Panel B of Table 7. In column (1), we also use cross-multiplier term of fintech and household income. Columns (2), (3), (4) and (5) are grouped regression according to total household income. Its results show that fintech can alleviate liquidity restrain to improve entrepreneurship. And the impact is only powerful for low-income family. It affirms inclusive impact of fintech and contribution to upward mobility.

At the enterprise level, poor households tend to start small and micro businesses. Therefore, we select micro and small enterprises as samples to study whether fintech can promote the development of micro and small enterprises. The data used comes from the 2015 China MSME Survey data from the China Household Finance Survey and Research Centre of Southwestern University of Finance and Economics (SWUFE). It samples MSMEs nationwide, and the samples are representative of the whole country, provinces, industries and key cities. Based on the cross-sectional data limitation, this article adopts the following regression equation to study the innovation of fintech on MSMEs.

$$\text{innovation}_{ip} = \beta_0 + \beta_1 x_p + \gamma c_j + \theta m_p + \varepsilon_i \quad (5a)$$

The explanatory variable "innovation" indicates whether a firm engages in innovative activities or holds a patent. It codes as 1 for yes and 0 for no. The core explanatory variable, "fintech," is lagged by one period at the provincial level. Control variables include firm-level and provincial-level variables. Detailly, we control the number of employees, years of managerial experience of the principal owner, and provincial-level economic development indicators, along with the share of education expenditure.

Regression results in Table 8 illustrate the impact of fintech on innovation across MSMEs of different assets. Column (1) presents results for the full sample, where the coefficient of fintech is significantly positive at the 10 % significance level. This indicates that fintech significantly enhances the innovation capacity of micro and small enterprises

Table 8
Small and micro businesses' innovation.

VARIVABLE	(1)	(2)	(3)	(4)
	Full sample	Less assets	More assets	Interaction terms
	Innovation	Innovation	Innovation	Innovation
Fintech	0.335* (0.195)	0.924*** (0.294)	0.029 (0.246)	4.488*** (1.548)
Fintech × ln(asset)				−0.286*** (0.105)
Observation	4317	1851	2674	3474
R-squared	0.0134	0.0237	0.00640	0.0479

Note: This table reports the results of whether fintech helps small and micro businesses' innovation. Column (1) uses full sample. We run grouped regressions based on the median assets as shown in Columns (2) and (3). Column (4) adds interaction terms based on column (1). All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

(MSMEs), thereby fostering intergenerational upward mobility.

Columns (2) and (3) further show results based on MSMEs categorized by the median of total assets. Columns (2) focuses on firms with fewer assets and Column (3) focuses on those with more assets. Fintech demonstrates a stronger and significant promotion of innovation among MSMEs with fewer assets, whereas its impact is not significant for MSMEs with more assets.

Column (4) includes a cross-multiplier term, revealing that fintech and firm assets jointly promote firms' innovation capabilities. Notably, fintech alleviates the constraints imposed by firm assets on innovation. The results underscore its role in easing liquidity constraints for MSMEs and boosting their innovation capabilities. This finding suggests that fintech supports MSME development, mitigates the risk of falling back into poverty, and facilitates upward mobility across generations.

6.2. Increasing the 'market luck' of the offspring

According to Becker and Tomes (1979, 1985), "market luck" represents factors unrelated to or unalterable by offspring and household characteristics, influenced by external macroeconomic factors. In our study, we define "market luck" as the fair opportunities available in the labor market.

Fintech has a dual role in boosting market luck. First, fintech's inclusivity is central to this process. By providing financial services to historically marginalized groups, it enhances access to capital for economically vulnerable segments (Hua and Huang, 2021; Molla and Biru, 2023). This is particularly significant in underdeveloped regions where financial services have traditionally been scarce. By enabling entrepreneurs from disadvantaged backgrounds to participate in the market, fintech fosters equal entrepreneurial opportunities and thus improves market luck.

Second, fintech harnesses internet technology to expand access to the digital economy. As the literature on online work and learning has shown, increased internet access helps reduce labor market inequality by opening doors to remote work, digital entrepreneurship, and online education. These opportunities are especially vital for individuals in marginalized communities who may otherwise face geographic or educational barriers. Research has demonstrated that internet access reduces labor market discrimination and increases economic opportu-

Table 9
Stimulating entrepreneurship in city-level.

	(1)	(2)	(3)
	Full sample	Underdeveloped places	Developed places
VARIABLE	Entrepreneurship	Entrepreneurship	Entrepreneurship
Fintech	0.036* (0.019)	0.080*** (0.027)	0.009 (0.026)
City-level controls	YES	YES	YES
City FE	YES	YES	YES
Year FE	YES	YES	YES
Observation	2515	1296	1219
Adjusted R-squared	0.958	0.922	0.970

Note: This table reports the results of whether fintech stimulates entrepreneurship in city-level. Column (1) uses full sample. Columns (2) and (3) are grouped regression according to urbanization rate. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Table 10

Type of newly created enterprises.

Panel A: Entrepreneurial enterprises of different natures		
VARIABLES	(1) Self-employed	(2) Private
Fintech	0.051** (0.023)	0.035* (0.019)
Observations	2515	2515
Adjusted R-squared	0.935	0.958

Panel B: Entrepreneurial enterprises with different factors of production					
VARIABLES	(1) Labor	(2) Capital	(3) Tech	(4) High-tech	(5) No-high-tech
Fintech	0.041* (0.022)	−0.026 (0.022)	0.046 (0.060)	−0.017 (0.039)	0.036* (0.019)
Observations	2515	2515	2515	2515	2515
Adjusted R-squared	0.945	0.962	0.956	0.966	0.958

Note: This table reports the results of whether fintech stimulates entrepreneurship of different types in city-level. In Panel A, we find the effects of the fintech on entrepreneurial enterprises with different factors of production. In Panel B, we try to study different aspects of fintech to influence entrepreneurship. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

nities across the board (Cowan, 2024; Dutcher and Saral, 2024; Makridis and Schloetzer, 2024; Nagler et al., 2024; Senik et al., 2024; Xu et al., 2024). Therefore, it improves market luck and enables intergenerational upward mobility.

6.2.1. Stimulating urban entrepreneurship

We utilize city-level business registration data obtained from the General Administration of Market Supervision website to gauge a city's entrepreneurial activity. Entrepreneurship is measured by the number of new business registrations. This data is integrated into a city panel dataset for examining fintech's impact on a city's entrepreneurship.

Additionally, we employ a group regression approach based on the median urbanization rate of cities in 2009. Cities below the median are categorized as non-developed regions, while those above are classified as developed regions. Regression results are presented in Table 9. Columns (1), (2), and (3) display results for the full sample, non-developed regions, and developed regions, respectively.

Under the full sample, fintech shows a positive coefficient at the 10 % significance level. In non-developed regions, fintech exhibits significantly positive coefficients at the 1 % significance level. Conversely,

Table 11

Increase internet exposure.

VARIABLE	(1)	(2)	(3)	(4)	(5)
	Work	Study	Business	Socialization	Entertainment
Fintech	1.090** (0.429)	1.692* (0.863)	1.295** (0.562)	0.004 (0.503)	0.897 (0.863)
Observation	2016	1944	2015	2015	2017
R-squared	0.148	0.341	0.268	0.182	0.265

Note: This table reports the results of whether fintech increases Internet exposure. The dependent variables in columns (1), (2), (3), (4) and (5) are Internet work, Internet learning, Internet business, Internet social and Internet entertainment, respectively. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

fintech does not yield significantly positive coefficients in developed regions. This suggests that fintech fosters greater entrepreneurial vigor in non-developed regions, effectively mitigating regional inequality. It stimulates economic growth, generates employment opportunities, and thereby promoting intergenerational upward mobility.

This study investigates the influence of fintech on newly established enterprises, categorizing them as self-employed or private, and further classifying them by factor intensity into labor-intensive, capital-intensive, and technology-intensive enterprises. Additionally, enterprises are distinguished based on their technological innovation, either high technology-intensive or non-high-technology-intensive.

The regression results in Table 10 demonstrate that fintech significantly affects the emergence of self-employed, private, labor-intensive, and non-high-technology-intensive enterprises. This finding corroborates earlier conclusions that fintech provides substantial benefits to low-income families and farmers. However, establishing capital-intensive, technology-intensive, and high technology-intensive enterprises poses challenges due to educational constraints. Nonetheless, these results underscore fintech's inclusive nature and its effectiveness in promoting intergenerational upward mobility.

6.2.2. Increase internet exposure

The data is from CFPS, which includes inquiries into Internet exposure. We examine various aspects of Internet engagement—work, study, business, socializing, and entertainment—as explanatory variables to assess fintech's impact. Regression results in Table 11 reveal significant

findings: fintech notably enhances the frequency of Internet work (significant at the 5 % level), Internet learning (significant at the 10 % level), and Internet business activities (significant at the 5 % level). However, fintech shows no significant effect on Internet socializing and entertainment frequencies.

These results underscore fintech's role in facilitating online work, business and study. A large literature has documented that Internet access reduces labor market inequality, particularly for online work and online learning (Cowan, 2024; Dutcher and Saral, 2024; Makridis and Schloetzer, 2024; Nagler et al., 2024; Senik et al., 2024; Xu et al., 2024). So fintech effectively boosts market luck by facilitating Internet access.

7. Heterogeneity analysis

To better understand the inclusive nature of fintech, we do some heterogeneity analysis. We present a heterogeneity analysis using grouped regression from two aspects: offspring and household. Detailly, we explore heterogeneity by offspring gender, offspring education, household property income and household income, respectively. The regression results are displayed in Table 12.

In Panel A of Table 12, we discover analysis by offspring gender heterogeneity. That's because there exists gender inequality. Traditional Chinese families prefer sons over daughters. So, we explored whether fintech could reduce this sexism and thereby achieve upward generational mobility. The results indicate that fintech significantly promotes absolute intergenerational upward mobility for female offspring at a 5 % significance level. However, it does not have a significant effect on the absolute intergenerational mobility of male offspring. This finding suggests that fintech narrows gender differences and reduces gender discrimination. This result is in line with the previous analysis of influence mechanism. Fintech alleviates parental liquidity constraints, so fathers can invest in girls' education. At the same time, fintech promotes Internet exposure to increase offspring market luck. This measure helps girls to reduce commuting stress and balance family and work through the Internet. So, it helps alleviate gender discrimination.

In Panels B and C, we explore the heterogeneity by household property income and total income. Since property income and total household income are the representatives of capital transmitted between generations. We want to see if fintech is inclusive and works for low-income families. The results prove that. Fintech is more conducive to households with no property income and low total household income. This is in line with our previous impact mechanism analysis. Fintech can enhance the investment of low-educated parents in their children and promote off-farm employment of farmers and entrepreneurship of low-income families. Thus, it proves inclusive nature of fintech again.

8. Conclusion

In the aftermath of the pandemic, both innovation and entrepreneurial activity have declined, while capital flows have encountered significant barriers. Therefore, advancing fintech development beyond just improving the financial system holds practical importance. This strategy aims to effectively promote inclusive finance by alleviating liquidity constraints and expanding market opportunities, particularly for disadvantaged groups such as the poor, small and micro businesses, and underdeveloped regions.

Our findings suggest that fintech significantly enhances absolute intergenerational mobility by reducing mobility constraints and improving "market luck." On one hand, fintech increases parental

Table 12
Heterogeneity analysis.

Panel A: Offspring sex heterogeneity		
	(1) Male URM _{50,30}	(2) Female URM _{50,30}
VARIABLE		
Fintech	0.003 (0.039)	0.051* (0.028)
Observation	1166	3107
R-squared	0.152	0.086
Panel B: Household property income heterogeneity		
	(1) No property income URM _{50,30}	(2) Have property income URM _{50,30}
VARIABLE		
Fintech	0.039* (0.020)	0.070 (0.094)
Observation	4068	586
R-squared	0.084	0.243
Panel C: Household income heterogeneity		
	(1) Low household income URM _{50,30}	(2) High household income URM _{50,30}
VARIABLE		
Fintech	0.065** (0.031)	0.034 (0.038)
Observation	2151	2522
R-squared	0.116	0.117

Note: This table reports the results of heterogeneity analysis. In Panels A, B, and C, we explore heterogeneity by offspring gender, household property income and household income, respectively. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

investment in children’s education and fosters entrepreneurship and innovation among low-income offspring. On the other hand, fintech boosts urban vitality and increases internet exposure, thereby reducing labor market inequality, which in turn improves the market prospects for younger generations. Moreover, the effect of fintech on intergenerational mobility varies due to its inclusive nature. Specifically, fintech helps reduce gender inequality and has a more substantial effect on households with no property income and low total household income.

Based on the conclusions of this study, we propose the following policy recommendations: First, promote the development of fintech and lower the entry barriers. The emergence of fintech has significantly reduced information asymmetry and streamlined loan procedures, thereby alleviating liquidity constraints and making financial services more accessible, particularly for low-income families. This has been instrumental in addressing the long-standing issues of "difficult and expensive loans." To further harness the potential of fintech, governments should take steps to promote its development by subsidizing the access of low-income households to fintech platforms. These subsidies could cover costs such as mobile devices, internet access, or transaction fees, which would make fintech services more accessible to disadvantaged groups. By doing so, governments can enhance the financial inclusion of low-income families, facilitating upward intergenerational

mobility and reducing economic disparities.

Second, address the potential negative impact of fintech. While fintech offers considerable benefits, it also presents potential risks, such as privacy concerns and digital exclusion. To mitigate these risks, it is crucial that governments introduce and enforce regulations to safeguard users’ rights and ensure responsible data usage. For instance, laws and regulations should be established to prevent the misuse of personal data and protect user privacy in fintech transactions. Additionally, to combat digital exclusion, policies should be enacted to ensure that marginalized populations, such as the elderly and those in rural areas, have access to the necessary infrastructure and digital literacy training to benefit from fintech services. Ensuring equitable access to these technologies will help prevent the deepening of existing inequalities.

Given data limitations, this study does not explore the full spectrum of fintech’s effects. Future research should further dissect the nuances of fintech, revealing additional channels through which it influences intergenerational mobility.

CRediT authorship contribution statement

Chenxuan Chen: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization.

Appendix

Table A1; Table A2 and Fig. A1

Table A1
Placebo test for IV results.

VARIVABLE	(1) <i>URM</i> _{50,30}	(2) <i>URM</i> _{50,0}
Topographic relief	0.001 (0.005)	0.012 (0.008)
Observation	24,329	24,329
Adjusted R-squared	0.00156	0.000651

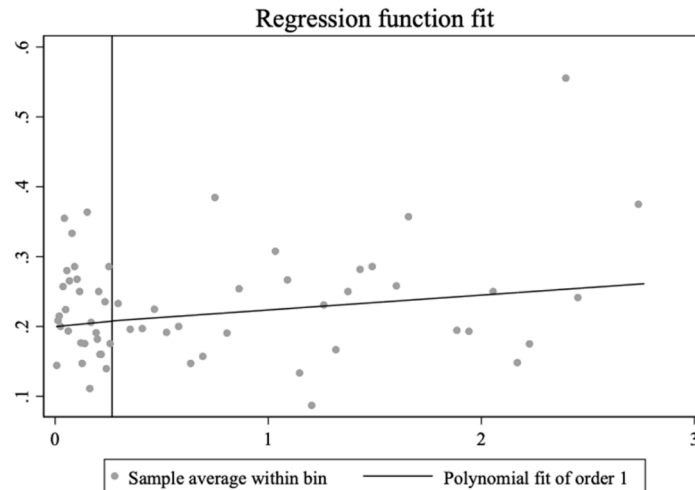
Note: This table reports the results of placebo test for IV results. We use data from China Household Income Project in 1988. The dependent variables in columns (1), and (2) are whether the offspring’s income rank is greater than 0 % and 30 % of the parent’s income rank when the parent’s income is ranked in the bottom 50 %, respectively. Since there is no more detailed geographic information, the explanatory variables are at the provincial level, and the control variables remain consistent with the baseline. Standard errors clustered at the province-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Table A2
Other IV results.

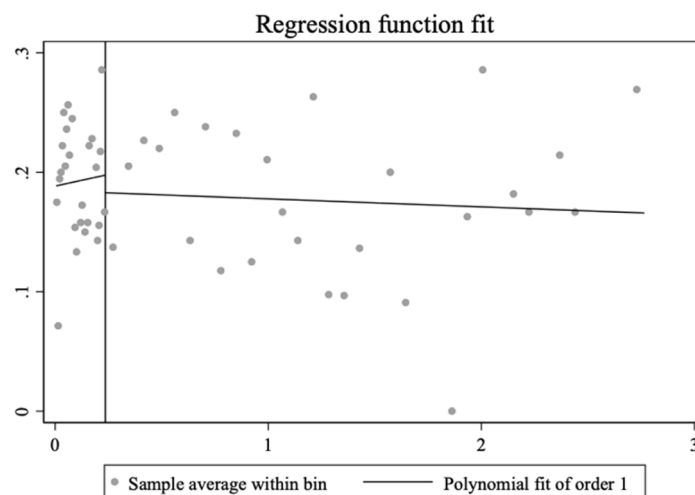
VARIVABLE	(1) Don’t control <i>URM</i> _{50,30}	(2) Add “Broadband China” <i>URM</i> _{50,30}
Fintech	0.142* (0.084)	0.208** (0.095)
Cragg-Donald Wald F statistic	524.33	218.04
Kleibergen-Paap Wald rk F statistic	29.90	11.29
Observation	4693	4693
Adjusted R-squared	0.040	0.037

Note: This table reports the results of other IV results. In Column (1), we use topographic relief as IV without controlling city-level variables. In Column (2), we use topographic relief and “Broadband China” policy as IV. All specifications control for year fixed effects and city fixed effects. Standard errors clustered at the city-level are in parentheses. ***, ** and * indicate p-value below 0.01, 0.05 and 0.1 respectively.

Panel A: Control group



Panel B: Experimental group

**Fig. A1. Exogeneity analysis.**

Note: This figure plots the exogeneity analysis for IV. Panel A is the results of control group, and panel B is the results of experimental group.

Data availability

The authors do not have permission to share data.

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